

Chapter 1

Chemistry

A science describes the matter - its properties – changes undergo

Energy

The capacity to do work or transfer heat

Law of conservation of matter

No change in matter quantity during chemical reaction or physical changes

Law of conservation of energy

Energy cannot be created or destroyed during chemical reaction or physical changes

Einstein equation (relationship between matter and energy)

$$E = mc^2$$

Law of conservation of matter and energy

The combined amount of matter and energy in universe is fixed (unchanged)

Chemical change

Occurs when with no change in a chemical composition

Physical change

Occurs when with no change in a chemical composition

Sublimation $\Rightarrow$ Endothermic

Conversion of solid directly into gas

Deposition $\Rightarrow$ Exothermic

Conversion of gas directly into solid

Evaporation $\Rightarrow$ Endothermic

Conversion of liquid directly into gas (or vapor)

Condensation $\Rightarrow$ Exothermic

Conversion of gas (or vapor) directly into liquid

Extensive properties

The properties depend on the amount of the material

Examples: Mass – Volume – Length - Weight

Intensive property $\Rightarrow$ Chemical properties

The properties independent on the amount of the material

Example: Color – Melting point- Density

Classify the below properties as extensive properties or intensive properties

boiling point, energy, volume, density, odor, concentration, mass

Extensive properties	Intensive property
mass	density
volume	boiling point
boiling point	odor
energy	concentration

Substance

It can not be broken down

Compound

A substance that can decomposed into simpler substances

Element

A substance that can not decomposed into simpler substances

Homogeneous Mixture

It has the same composition, components are indistinguishable

Heterogeneous Mixture

It does not have the same composition, components are distinguishable

Molecule

It is two or more atoms joined tightly and behaves as single

Chemical equation

It is used to represent the chemical change

Symbol

It is used to represent the element

Law of conservation of mass

A total mass of the substances does not change

The total mass of the products is equal to the mass of the starting reactants

Law of definite proportion

Elements are combined to form compound in the fixed proportion by mass

Dalton theory

Elements compose from small indivisible particles called atom (Billiard balls)

Bohr theory

Electrons circling the nucleus (planet orbits)

Modern atomic theory

Electrons make cloud region around nucleus

Rules:

Electrons assign to orbit in order of increasing $(n + 1)$

A maximum number of electrons in orbit is given by $(2n^2)$

Paulis exclusion principle:

No two electron can have the same (identical) four quantum numbers

Hund rule

The electrons occupy orbits singly before pairing